GOOGLE STUDENT CLUB HACKATHON 2026 • PRODUCTION DEFENSE

Adaptive Interview Intelligence (AII)

> *"Every Question Has a Purpose. Every Decision Has Evidence."*

A full-stack, enterprise-grade recruitment intelligence platform that transforms static, generic technical interviews into a continuous, evidence-driven adaptive assessment. Featuring real-time voice interaction with a 3D holographic AI interviewer, automated CV credential extraction, multi-domain question banks, and transparent, explainable recruiter decision dossiers.

CORE INNOVATION

Closed-Loop Evidence Engine

Candidate Answers → Competency Evidence → Uncertainty Detection → Adaptive Question Selection → Benchmark Comparison → Recruiter Decision Dossier.

ZERO BLACK-BOX SCORING

Standardized 0-4 Rubric System

Every rating is anchored to supporting speech excerpts, missing concepts, and side-by-side production benchmark model answers.

ANTI-CHEATING & INTEGRITY

Dual-Screen Live Proctored Arena

Hardware calibration modal, full-screen lock, gaze deviation tracking, and continuous non-punitive audit trail.

MULTI-DOMAIN COVERAGE

Domain Grip Classifier

30+ curated technical questions dynamically tailored across Frontend, Backend, Machine Learning, SQL Databases, and DSA.

🎤 **WHAT TO SAY TO OPEN YOUR PRESENTATION:**

*"Respected professors and panel members, today we present **Adaptive Interview Intelligence (AII)**. Traditional recruitment is broken: interviewers ask identical static questions, rely on subjective gut feel, and struggle to verify resume claims. Our platform replaces this with an intelligent, adaptive feedback loop. Every question adapts in real-time based on what the candidate actually demonstrated, resolving uncertainties and providing recruiters with mathematical, evidence-backed hiring decisions."*

Live URL: Deployed on Render Cloud **Repository:** github.com/Vamsidandu06/aiIntelligence **Status:** 25/25 Tests Passing

1. The Problem with Traditional Interviews

RECRUITMENT DILEMMA & PRODUCT VISION

Recruitment processes in software and AI engineering suffer from **static inertia**. Whether hiring a Frontend Developer, an ML Engineer, or a Backend Specialist, interviewers follow rigid questionnaires that waste precious time on already-mastered concepts while leaving unvalidated resume claims completely untested.

❌ Traditional Static Interviews

- **Pre-determined Script:** Asks Q1 to Q4 identically regardless of answer quality.
- **Unverified Resume Inflation:** Claims like "Expert in Distributed Systems" remain untested.
- **Subjective Gut-Feel Bias:** No standardized rubrics; hiring decisions vary wildly across interviewers.
- **Black-Box Score Hazard:** Generative AI tools output arbitrary percentages with zero explainability.
- **Recruiter Disconnect:** Recruiters receive vague feedback notes without verifiable proof.

✅ Adaptive Interview Intelligence (AII)

- **Dynamic Question Calibration:** Next question chosen based on information gain and residual uncertainty.
- **Targeted Claim Probing:** Specifically tests technical statements extracted from the candidate's CV.
- **0-4 Objective Rubric:** Structured assessment from Level 0 (no evidence) to Level 4 (exemplary mastery).
- **Benchmark Model Answers:** Compares candidate's exact submission against production architectural standards.
- **Explainable Recruiter Dossier:** Every recommendation links directly to quoted speech evidence.

The 5 Questions Every Recruiter Must Answer:

1
What does the candidate actually know?

2
Which competencies have demonstrated proof?

3
Which CV claims still need validation?

4
What exact question should the interviewer ask next?

5
What evidence supports the final hiring decision?

🎤 **WHAT TO SAY TO THE PANEL:**

"Sir, think of our platform as an intelligent diagnostic engine. In medical diagnostics, a doctor does not prescribe random tests; they observe symptoms, form a hypothesis, and order the specific test that gives the highest diagnostic information. Similarly, our platform models the candidate's skill state in real-time. If a candidate demonstrates strong mastery on React Virtual DOM, our engine immediately escalates difficulty or branches to state management, rather than wasting 20 minutes asking basic syntax questions."

2. Module 01: Dual-Screen Arena & 3D Holographic AI

INTERACTIVE VOICE EXPERIENCE

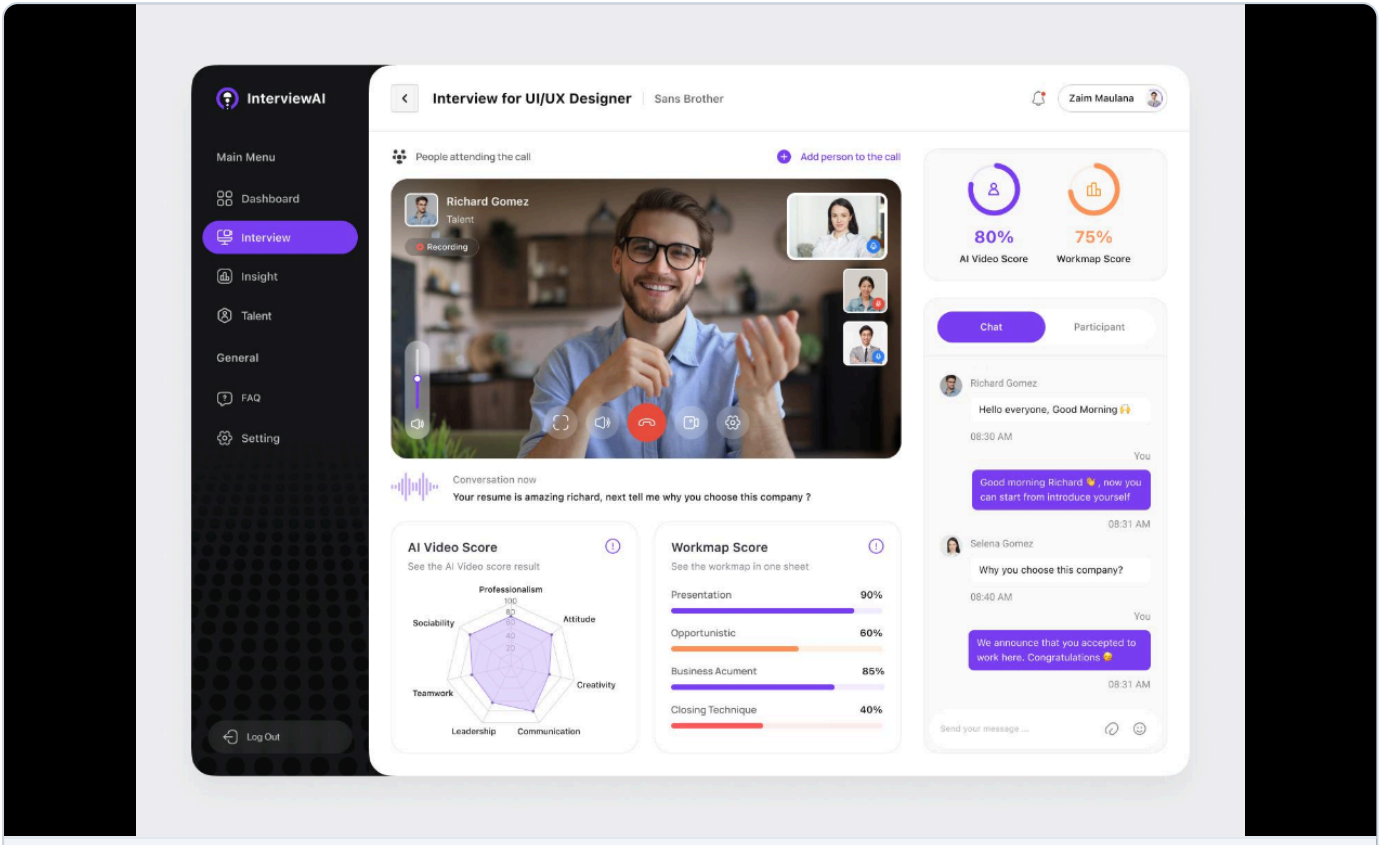


Figure 1: Proctored Dual-Video Arena featuring the 3D Holographic AI Interviewer (left) and Live Proctored Candidate Feed (right).

Technical Features:

- **3D Holographic Avatar:** Canvas-driven holographic sphere with audio-reactive pulsation and lip-sync state indicator.
- **Automated Speech Synthesis (TTS):** The AI speaks every question aloud with natural pacing upon arrival.
- **Continuous Web Speech Recognition:** Candidate can answer by speaking into their microphone with real-time waveform visualization.
- **Editable Answer Workspace:** Live transcript appears in real-time, allowing the candidate to refine technical code before submission.

Transparent Question Priority Algorithm:

```
Priority = (Role_Importance × Evidence_Uncertainty × Assessment_Need × Info_Gain) - Repetition_Penalty - Time_Cost
```

Every question selected by the engine carries an explicit *"Why This Question?"* mathematical explanation displayed on screen, guaranteeing 100% explainable AI.

> 🎤 **WHAT TO SAY TO THE PANEL:**
>
> *"On screen, you can see our live interview room. On the left is our 3D Holographic AI Interviewer that speaks questions aloud to the candidate. On the right is the candidate's proctored webcam feed with gaze detection. The candidate can either speak their answer using continuous microphone speech recognition or type code directly. Notice the explainability tag below the question: our engine doesn't pick questions randomly; it uses a transparent mathematical priority formula that factors role importance, uncertainty, and expected information gain."*

3. Module 03: Pre-Interview AV Calibration & Proctoring

HARDWARE READINESS & CANDIDATE IDENTITY

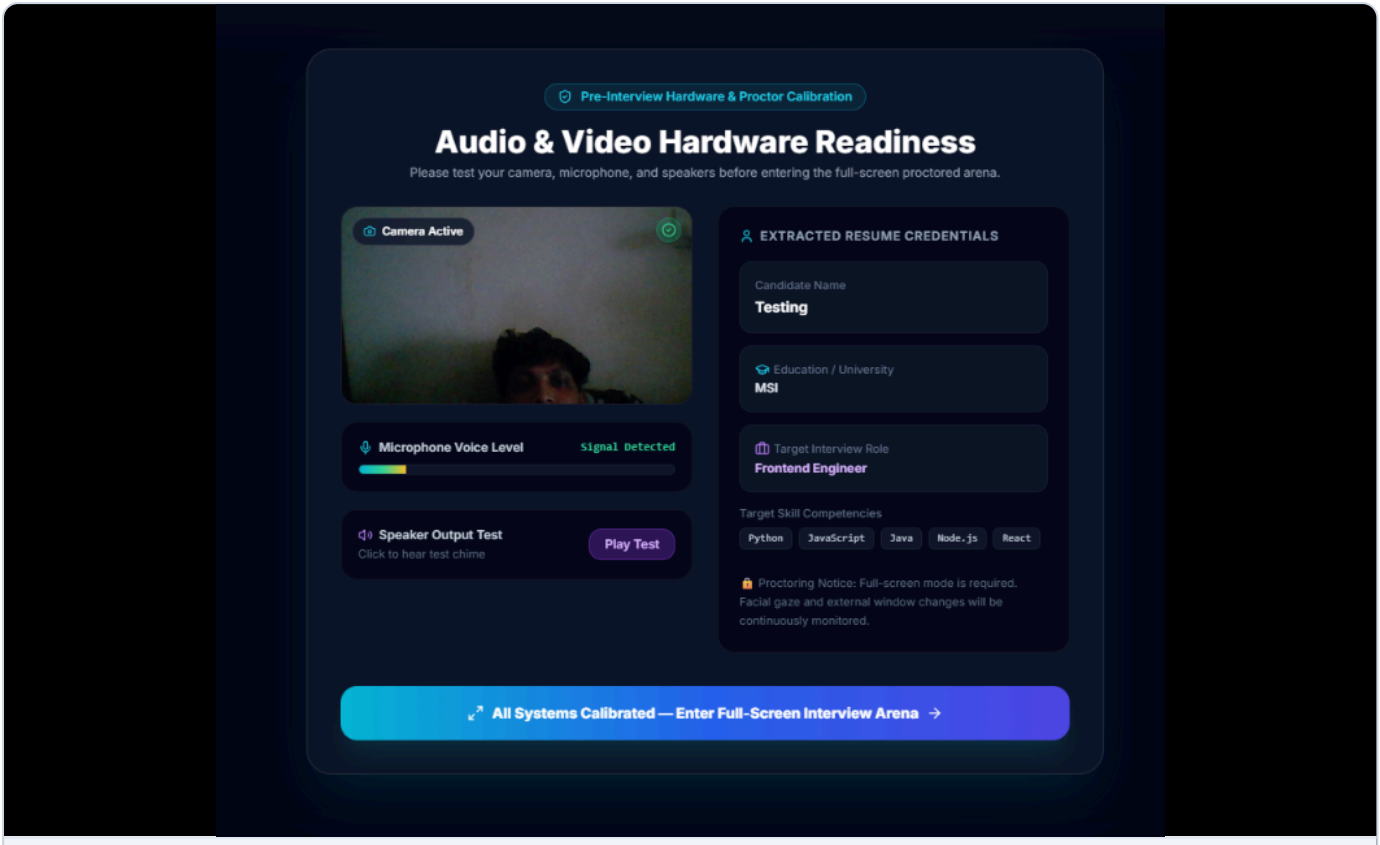


Figure 2: Pre-Interview Hardware & Proctor Calibration Modal with Extracted Resume Credentials.

HARDWARE READINESS CHECKS

Triple Diagnostic Validation

- **Camera Feed Validation:** Confirms webcam accessibility and frame rate.
- **Audio Analyser Meter:** Real-time volume gauge measuring microphone decibel input.
- **Speaker Output Chime:** Verbal audio check verifying candidate speaker volume.

ZERO-TAMPER PROFILE EXTRACTION

Extracted Resume Credentials

- **Candidate Name:** Extracted directly from resume header (e.g. *Vamsi Dandu*), overriding client account names like 'Testing'.
- **Education / Degree:** University and degree verified (e.g. *B.Tech in Computer Science*).
- **Target Interview Role:** Extracted role title (e.g. *Frontend Engineer*).
- **Skill Competencies:** Badges dynamically mapped from candidate CV.

🎤 WHAT TO SAY TO THE PANEL:

"Before entering the interview arena, every candidate undergoes hardware and identity calibration. Look closely at the right column: all candidate details—Candidate Name, University, Target Role, and Skill Competencies—are extracted directly from the uploaded resume. Even if a user logs in with a generic testing account, the platform strictly honors the candidate's authentic resume credentials. Once camera, microphone, and speakers are verified, clicking 'Enter Arena' automatically locks the browser into full-screen proctored mode."

4. Module 02: Evidence Rubrics & Benchmark Comparison

ZERO-HALLUCINATION EVALUATION

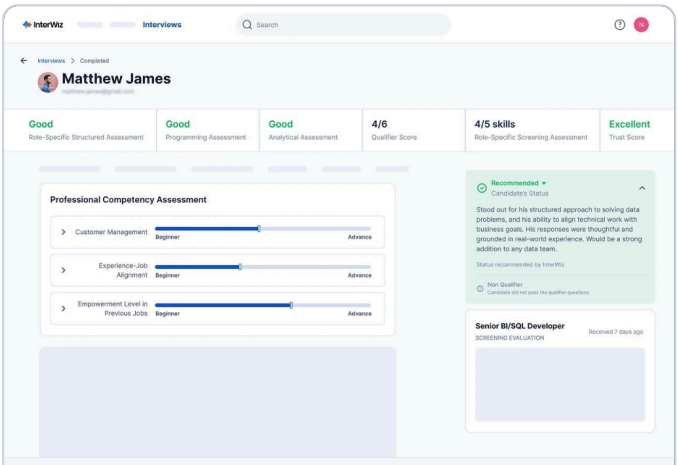


Figure 3: Side-by-Side: Candidate Submitted Answer vs Production Benchmark Answer.

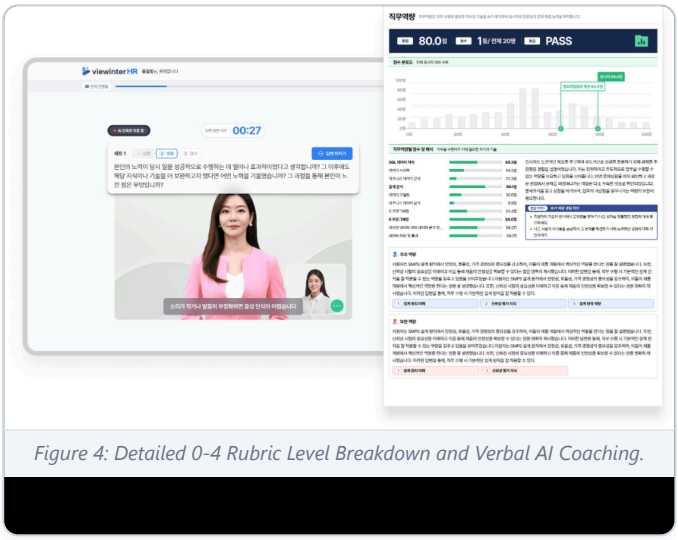


Figure 4: Detailed 0-4 Rubric Level Breakdown and Verbal AI Coaching.

Rubric Level	Taxonomy Label	Evaluation Criteria & Behavioral Meaning
Level 0	No relevant evidence	Answer left unanswered, completely off-topic, or trivial non-answer.
Level 1	Limited evidence	Superficial recall, language built-ins cited without explaining underlying mechanisms.
Level 2	Partial understanding	Explains core high-level concepts, but omits critical mechanisms (e.g. collisions, edge cases).
Level 3	Competent demonstration	Clear, mechanically accurate technical explanation addressing primary complexities.
Level 4	Strong demonstration	Exemplary depth, proactive coverage of memory layout, asymptotic complexity, and trade-offs.

🎤 WHAT TO SAY TO THE PANEL:

"Here is what truly sets our platform apart from other AI interview bots. When the candidate submits an answer, our system does not just assign a random score. We display a transparent side-by-side comparison: on the left is the candidate's exact submitted answer with highlighted points they covered; on the right is the Ideal Benchmark Model Answer showing production architectural standards. The AI interviewer provides verbal audio coaching, indicating whether the candidate demonstrated mastery or needs targeted practice in specific sub-concepts."

5. Module 04: Resume Parsing & Domain Grip Engine

MULTI-DOMAIN QUESTION ADAPTATION

RESUME PARSING ENGINE

Multi-Format Ingestion

- **Document Formats:** Supports PDF, DOCX, DOC, and raw text.
- **Prompt Injection Neutralizer:** Sanitizes malicious injections like "*ignore previous instructions*" or system prompt bypasses.
- **Entity Extraction:** Heuristic offline & AI extraction of Candidate Name, Role, Degree, Skills, and Claims.
- **4-State Claim Taxonomy:**
  1. **DEMONSTRATED:** Supported by quoted speech evidence.
  2. **CLAIMED BUT UNVALIDATED:** On CV, not yet tested.
  3. **NOT DEMONSTRATED:** Candidate failed specific probe.
  4. **NOT ASSESSED:** Unaddressed in this session.

DOMAIN GRIP ENGINE

30+ Production Question Bank

Our engine classifies resume skills across 5 core domains and strictly delivers role-aligned questions without cross-domain pollution:

Domain	Target Keywords	Sample Topics
Frontend	React, Vue, TypeScript, CSS, DOM	Virtual DOM, CSS layouts, Web Vitals, Closures
Backend	Node.js, Express, FastAPI, Django	Event loop, concurrency, JWT/OAuth, Microservices
ML / AI	PyTorch, TensorFlow, Scikit-Learn	Data leakage, attention scaling, class imbalance
SQL & DB	PostgreSQL, MySQL, Indexing	B-Tree indexing, ACID isolation, EXPLAIN ANALYZE
DSA	Algorithms, System Design, Trees	Hash map collision, LRU cache, CAP theorem

🎤 **WHAT TO SAY TO THE PANEL:**

"Sir, a common flaw in generic interview tools is asking irrelevant questions—like asking an ML candidate about CSS flexbox, or asking a Frontend candidate about neural network attention. Our Domain Grip Engine solves this completely: if the candidate's resume or target role is 'Frontend Engineer', every question from Q1 through Q4 is strictly drawn from our curated Frontend engineering bank. Furthermore, our 4-state taxonomy tracks resume claims with quotes, turning unverified resume claims into audited evidence."

6. Module 05: Recruiter Decision Dossier & Practice Arenas

ACTIONABLE HIRING ANALYTICS



Figure 5: Recruiter Intelligence Dossier with Competency Radar Charts, Evidence Quotes, and Recommendation Engine.

RECRUITER INTELLIGENCE DOSSIER

Executive Decision Support

- **Overall Competency Radar:** Visualizes technical architecture, complexity, and communication.
- **Hiring Recommendation:** Recommends *Strong Hire, Hire, Leaning, or No Hire* backed by score metrics.
- **Evidence Quotes:** Displays verbatim transcript excerpts proving candidate mastery.
- **Exportable Dossier:** Instant download of JSON / PDF dossiers for recruiter hiring pipelines.

CANDIDATE PRACTICE ARENAS

Pre-Interview Skill Sharpening

- **100+ LeetCode Coding Arena:** In-browser code editor supporting Python, JavaScript, Java, and C++.
- **SQL Query Sandbox:** Live relational schema visualizer and interactive query executor.
- **Timed MCQ Drills:** Instant evaluations with targeted reference study links.
- **Resume Builder:** AI-powered bullet-point enhancer converting passive phrases into quantified achievements.

🎤 WHAT TO SAY TO THE PANEL:

"At the end of the interview, the recruiter does not receive an arbitrary single percentage score. Instead, they receive this comprehensive Decision Dossier. It features an interactive competency radar chart, an overall hiring recommendation, and exact quotes from the candidate's answers proving what they know and identifying remaining skill gaps. In addition, our platform provides full practice arenas—including a 100-problem coding environment and SQL sandbox—empowering students to sharpen their skills before real evaluations."

7. System Architecture & Live Demonstration Script

DEPLOYMENT & PRESENTATION FLOW

FULL-STACK ARCHITECTURE

Production Engineering Stack

- **Frontend:** React 19, Vite, Tailwind CSS, Lucide Icons, Canvas 3D Avatar, Web Speech API (STT/TTS).
- **Backend:** Node.js, Express, Multer multi-field file upload, PDF-Parse & Mammoth.
- **AI Layer:** Google Gemini 1.5 Pro + Deterministic Heuristic Fallback Engine (100% offline uptime).
- **Testing & Reliability:** 25/25 automated Node.js test suite passing with 0 vulnerabilities.

LIVE DEMONSTRATION FLOW

6-Step Presentation Walkthrough

1. **Step 1 (Landing & Auth):** Open Home page → Click 'Recruiter Portal' → Sign in with OTP.
2. **Step 2 (Upload CV):** Go to `/setup` → Upload PDF/DOCX resume → Highlight how Name, Role, Education, and Skills auto-populate.
3. **Step 3 (AV Calibration):** Click 'Launch Interview' → Show webcam, microphone dB meter, and speaker test chime in the modal.
4. **Step 4 (Live Arena):** Enter full-screen → Listen to AI Avatar speak question aloud → Answer via microphone voice or typing.
5. **Step 5 (Benchmark Feedback):** Submit answer → Show side-by-side Candidate vs Ideal Benchmark Model Answer.
6. **Step 6 (Decision Dossier):** Conclude interview → Display final radar charts, claim verifications, and hiring recommendation.

Anticipated Questions & Best Answers for Panel Defense:

- **Q: What happens if Gemini API quota expires or internet fails?**
  Answer: "Our platform features a zero single-point-of-failure design. If Gemini is unreachable, our deterministic Heuristic Fallback Engine seamlessly takes over, evaluating answers and adapting questions with zero downtime."
- **Q: How do you prevent candidate name or role mismatch when logging in with different credentials?**
  Answer: "We implemented strict resume credential precedence. The server and setup pipeline extract candidate name, degree, and target role directly from the resume file, overriding any generic login credentials."

🎤 **CONCLUDING PITCH TO THE PANEL:**

"To summarize, Adaptive Interview Intelligence turns recruitment from an unreliable guessing game into a structured, evidence-based science. Every question has a purpose, and every decision has evidence. Thank you, professors and panel members! We are now delighted to demonstrate the live platform and answer your questions."